Modelling and Simulation

Lecture 1: Introduction and general concepts

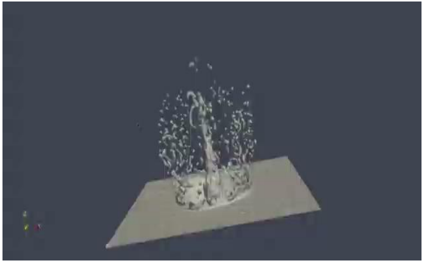
1. Objectives and Background

- Learn how to describe natural phenomena
- Give an introduction to several modelling techniques
- Show how these models can be simulate on a computer (in-silico laboratory, numerical experiments)
- Learn more about science through modelling

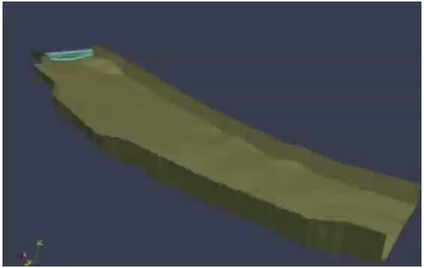
What natural processes are we interested in ?

- Physics (Fluid mechanics), astrophysics, chemistry, climatology,...
- Environmental sciences (river modelling, Volcano plume)
- Biology (Tissue growth), pattern on animal skins, cells, organs
- Ecosystems: competition between species, ant behavior, equilibrium between forest and savanna, propagation of epidemia, …
- Finance, social sciences, traffic, pedestrian motion, …
- …

Movies

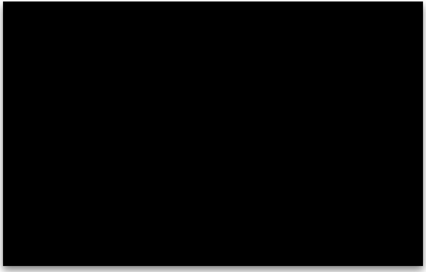


Droplet



River

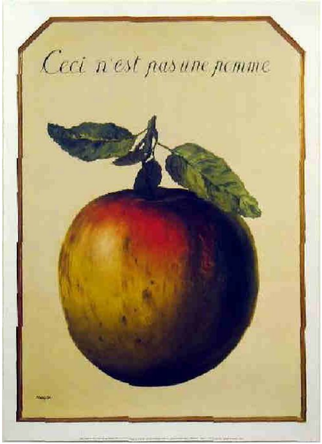
Movies



Volcano

River

What is a model?



- This is not an apple, just its graphical representation

What is a model?

Many possible definitions:

- Simplified abstraction of reality, allowing us to better describe and understand it
- An abstraction in which only the essential ingredients are retained, according to the question we ask about the system
- It is the representation of a phenomena in a mathematical or computer-based language

Computational Science

Many skills are needed to build a new model, to run it and analyze its results.

- Computational Science is an emerging, multidisciplinary domain, based on the idea of **“computational thinking”**
- A computer-based description offers a new language, a new methodology to address scientific challenges, far beyond the scope of traditional numerical methods and in fields where these classical approaches hardly apply.

Computational Science

- **Modelling and simulation** is a central part of computational sciences. It is a response to the new questions scientists want to solve, resulting from the avalanche of new experimental data, and the need to integrate many processes together rather than specializing in one single problem.
- A computational scientist needs to be a physicist, a mathematician, a computer scientist, a biologist, an economist, …

2. Modelling and Simulation

Why a model?

- Describe, classify, but mostly
- Understand
- Predict
- Control a phenomena

What is a good model?

It depends on the question. Several models may be necessary for studying different aspects of the same phenomena.

Everything should be made as simple as possible but not simpler.

Albert Einstein

Level of reality

The same system can be described at different scales, and different methods apply depending on the scale of interest:

- atoms, molecules, fluid elements, pressure field, climat
- cells, tissues, organs, living being
- mechanical parts, cars, traffic

Level of reality

- One has to identify the important ingredients and their mutual interactions
- Often, one defined a model at finer scale than the scale at which we ask a question

Several models / different language of description

Partial differential equation for a fluid:

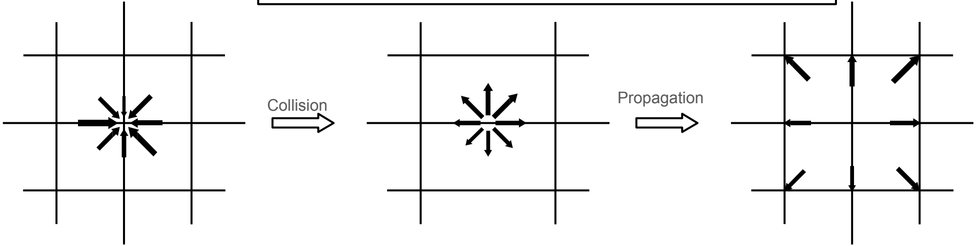
$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \mathbf{u}$$

phenomena → PDE → discretisation → numerical solution

… to a virtual model of reality

One considers a discrete universe as an abstraction of the real word

phenomena → computer model



- Mesoscopic *Rule* describing the phenomena

Example of modelling method

- N-body systems, molecular dynamics
- Mathematical equations, ODE, PDE
- Monte-Carlo methods (equilibrium, dynamic, kinetic)
- Cellular Automata and Lattice Boltzmann methods
- Multi-agents systems
- Discrete Events simulation
- Complex networks

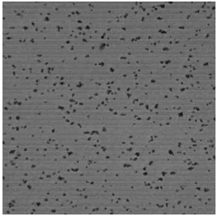
From a model to a simulation

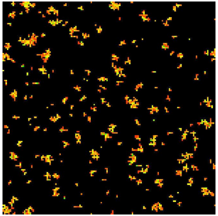
- Once a model is specified, one need to program it, to run it (many times) and to study the results
- It is a numerical experiment in **computer based virtual universe**
- One need to understand computer programs, software engineering, algorithms, data-structures, hardware (parallel machines, GPUs), code optimization, data-analysis.

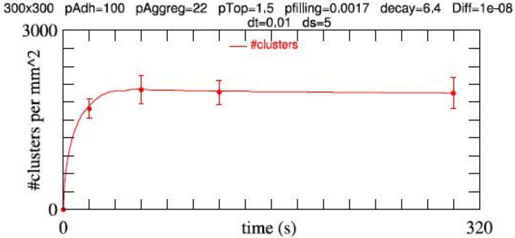
From a model to a simulation

- The program needs to be verified (did we really implemented the model ?)
- The model should be validated (run benchmarks with known results).
- One need enough knowledge of the phenomena to judge if its predictions are acceptable in new situations.

From a model to a simulation







3. Modelling Space and Time

Space and time

- Natural processes occur in space and evolve over time (spatially extend dynamical systems)
- For instance the atmospheric temperature is different from one place to another, and changes over time
- Also, a car on a road changed position as time goes on
- Sometime one is only interested in the time evolution of a quantity, regardless of the spatial location (e.g. the number of individuals) in a population
- Sometime, a process is stationary (no time evolution). Then only the spatial variations are of interest (e.g temperature in a room, in the middle or near the windows).

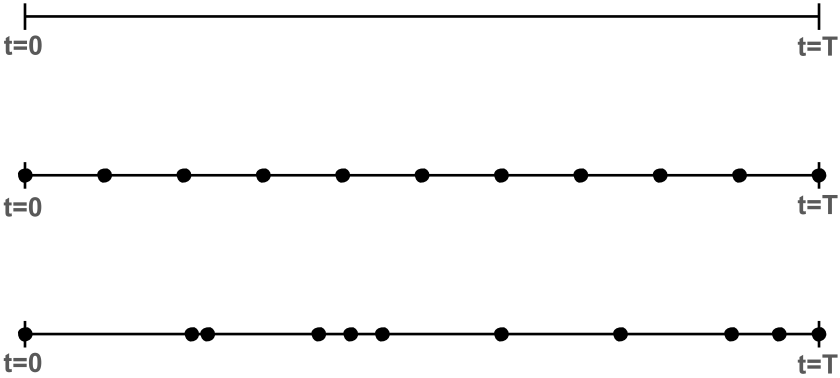
Time evolution

- To capture the temporal dimensions in a model, there are several ways:
  - Time takes any real values (as physics suggests). Only mathematical models can deal with this approach (differential equations)
  - Otherwise, the duration of the process is broken up in small time intervals Δt and one describes the state of the system at each of these **time-step** $t_0=0$, $t_1=\Delta t, \ldots, t_n=n\Delta t \ldots$
  - the time is discretized, but the process is followed **continuously** over its duration.

Time evolution

- Alternatively, we can focus on the interesting moments of a process
- In a queue in front of a post office booth, one can simply consider the time at which a remarkable event occurs. For instance a new customer enters or a previous one is done.
- The time t at which an event occurs can be any real value
- The time is not discretized but the evolution of the system is broken up according to events
- This is so-called Discrete-Event-Simulation (DES) approach

Time evolution



Modelling space: Eulerian approach

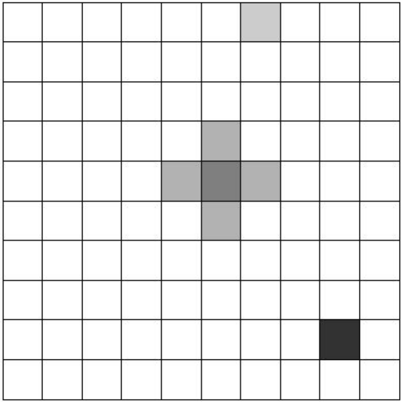
To include the spatial dimensions in a model, there are also different ways.

- One can take the point of view of an observer who sits at a fixed position $\vec{x}$ in space and records what he sees.
- For instance the local atmospheric pressure $p(\vec{x}, t)$
- Or the number of cars that passed be every minute
- This is so-called **Eulerian** approach: attach a property of the system at each spatial locations.
- Space can be continuous (mathematical models) or discretized in cells, forming a mesh covering the region of interest.

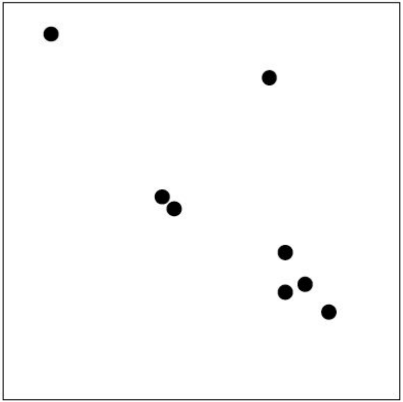
Modelling space: Lagrangian approach

- Alternatively, one can give the position of all the objects of interest, as a function of time.
- For instance the movement of the Moon is described by its trajectory $\vec{x}(t)$, where $\vec{x}$ is a continuous variable.
- In a traffic model, one can give the position over time of all the cars.
- This is so-called **Lagrangian** approach: the observer take the point of view of the moving objects.

Modelling space



Eulerian point of view



Lagrangian point of view

Beyond the physical space: complex networks

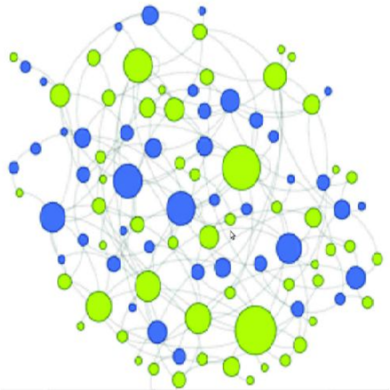
- In many systems, it is not so much the exact spatial positions of the components of a system that matters
- It is rather whether these components see each others, or can interact
- This is typically the case in social systems. Two persons can be very far away but still interact a lot by phone or other means

Beyond the physical space: complex networks

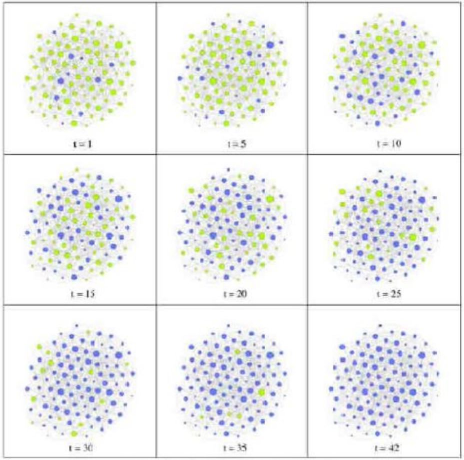
- For instance, the agent in an economical model can be represented as a **graph** (or a complex network): an edge connects pairs of agents that exchange information, money, goods,...
- Obviously, such a graph can be dynamical: creation of new links or destruction of old ones.

Example

A model of opinion propagation in a social network



(Lino Valasquez, UNIGE)



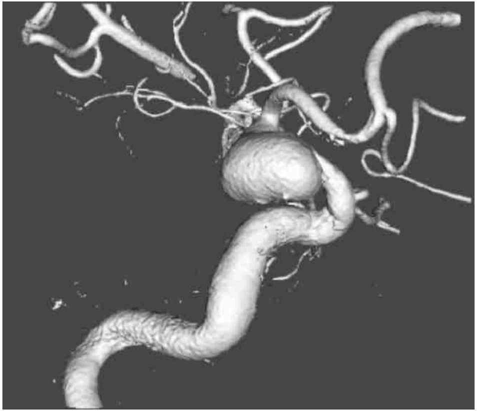
Réseau aléatoire, N = 100, p = 0.05, ε = 0.5

Complex networks

- Dynamical systems on complex networks is a fast developing field
- Graph topology imposes a rich “spatial” structure which constraints the dynamics
- Many quantities characterize the graph topology and can be related to some global properties of the system: degree distribution, clustering coefficient, centrality measures, assortativity, etc.

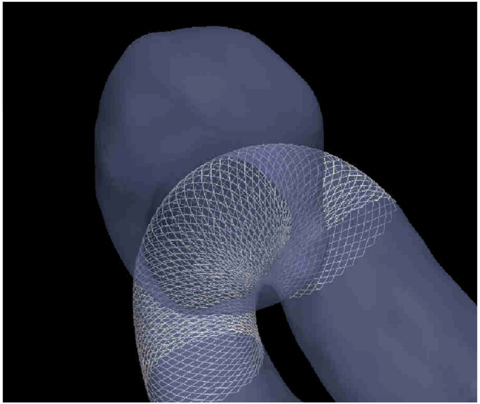
4. Example of bio-medical Modelling

Thrombosis in cerebral aneurysms



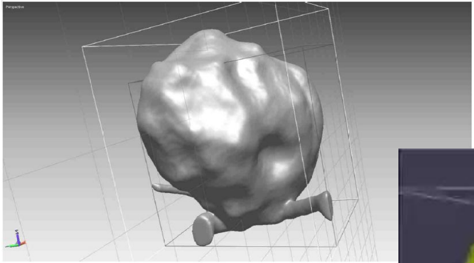
The evolution of aneurysms is driven by bloodflow, biomechanics and biology.

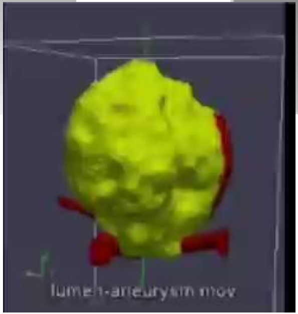
How to treat them: Stents/flow diverter

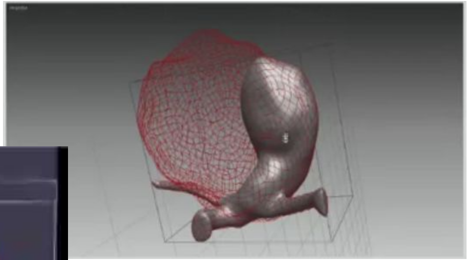


- the flow-diverter **reduces bloodflow** in the aneurysm
- **Clotting is induced in the aneurysm**

Example of a thrombus in a segmented aneurysms



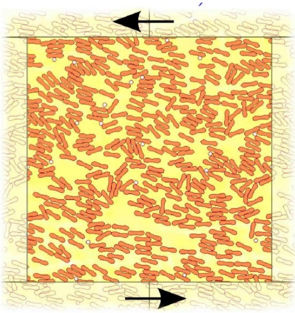




(Image processing:
Guy Courbebaisse,
INSA-Lyon)

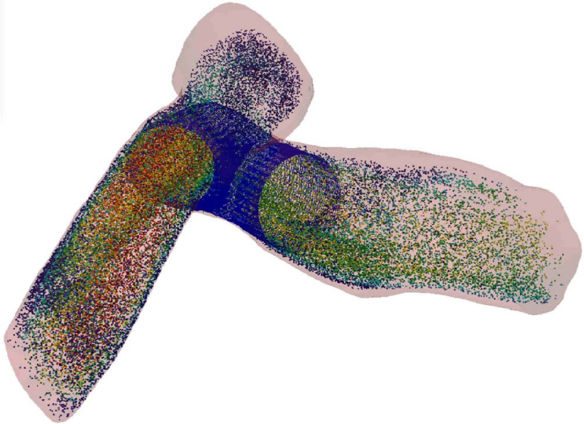
Micro-model (University of Amsterdam)

Red Blood Cells and platelets are modeled as deformable objects in suspension in the plasma fluid



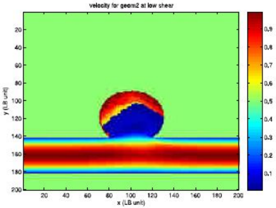
[Platelets and RBC in a 2d shear](#)
[3D suspension model](#)
(L. Mountrakis, E Lorenz and A.G. Hoekstra)

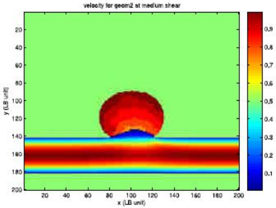
Macroscopic flow simulation in an aneurysm

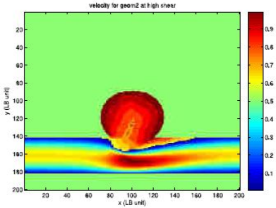


Numerical Model of Clotting

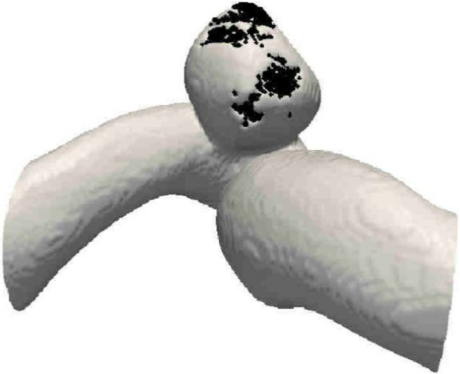
Under proper flow conditions and depending on the vessel wall biological response, a clot develops: a solidification of the blood from the wall.







Numerical Model of Clotting



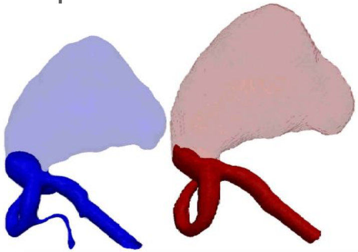
Pulsatile versus steady flow.

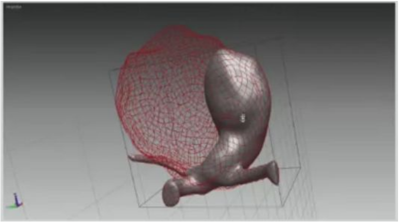
Qualitative validation

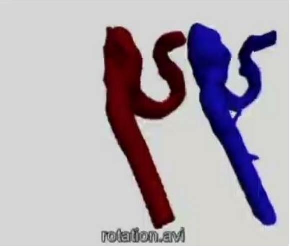
As observed in giant aneurysms, the simulation shows:

- Partial or total aneurysm thrombosis
- Spontaneous stop and start of the clot

Blue: patient   Red: simulation







5. Monte-Carlo methods I

Background

- The goal of Monte-Carlo methods is the sampling of a process in order to determine some statistical properties
- For instance, we toss a coin 4 times. What is the probability to obtain 3 tail and 1 head?
- Mathematics gives us the solution:

$$P(3head) = \binom{4}{3}\left(\frac{1}{2}\right)^3\left(1-\frac{1}{2}\right)^1 = \frac{1}{4}$$

- But we could also do a simulation

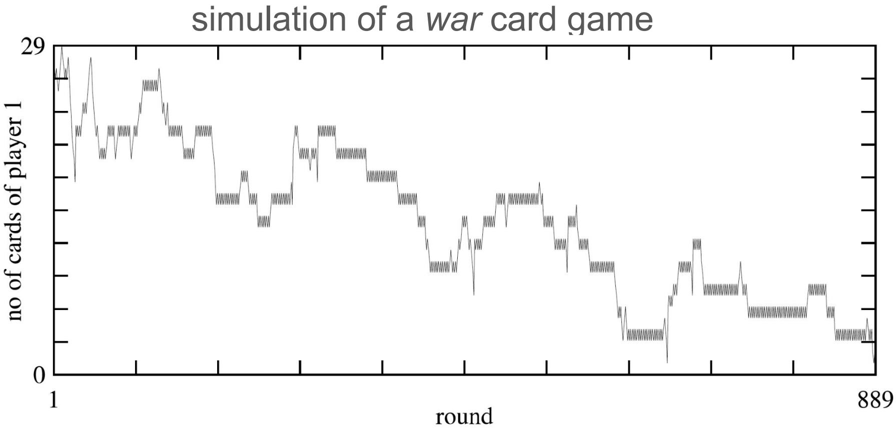
A Monte-Carlo computer simulation

```
import random

def monte_carlo_coin_toss (num_simulations):

 successful_outcomes = 0
 for _ in range(num_simulations):
   tosses = [random.choice([ 'H', 'T']) for _ in range(4)]
   if tosses.count('T') == 3 and tosses.count('H') == 1:
     successful_outcomes += 1

 return successful_outcomes / num_simulations

# Run the simulation with a large number of trials
num_simulations = 100000
estimated_probability = monte_carlo_coin_toss(num_simulations)

print(f"Estimated probability of getting 3 tails and 1 head in 4 coin tosses:
{estimated_probability:.4f}")
```

More difficult problem

- For the coin tossing problem, no need for a simulation
- But we can think of other problems for which probability theory could hardly applied
- For instance: what is the average duration of the card game called “war” (or battle)?

The *war* card game with 52 cards



Historical note

- The method was name in 1940s by **John von Neumann**, **Stanislaw Ulam** and **Nicholas Metropolis** after the name of the *Monte-Carlo casino*, where Ulam's uncle used to gamble … and lose his money
- The motivation was to find out the probability than a Canfield solitaire will finish successfully
- Ulam found it easier to play many Canfield solitaires and estimates the number of successes rather than trying ro apply combinatorics and probability theory
- Then the Monte-Carlo method was successfully applied to the *Manhattan project* (nuclear weapon) in the Los Alamos National Laboratory

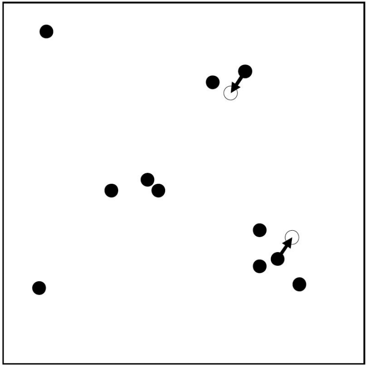
6. Monte-Carlo methods II

Markov-Chain Monte-Carlo (MCMC)

- We consider a stochastic process whose goal is to explore the state space of a system of interest
- Let x be a point in this state space. Let us assume that this point modes across the space by jumping randomly to another point x'.
- The jump from location x to location x' takes place with probability $W_{x \to x'}$. This advanced the system time from t to $(t+1)$. (Markov chain).

Markov-Chain Monte-Carlo (MCMC)

- We want this process to sample a prescribed probability $\rho(t, x)$. This stochastic process should be at point x at time t with a probability $\rho(t, x)$.
- How do we choose $W_{x \to x'}$?



$$\rho \propto \exp(-E(x)/k_B T)$$

Sampling the diffusion equation in 1D

The probability that our random exploration process is it at location x at time $t+1$ is

$$p(t+1) = \sum_{x'} p(t,x') W_{x' \to x}$$

Let us consider a 1D discrete space: $x \in \mathbf{Z}$

where one can move to the right with probability ***W₊*** , to the left with probability ***W₋*** and stay still with the probability ***W₀***.

The equation for $p(t,x)$ simplifies to

$$p(t+1,x) = p(t,x-1)W_+ + p(t,x)W_0 + p(t,x+1)W_-$$

Diffusion equation (1D)

The diffusion equation is $\delta_t \rho = D \delta_x^2 \rho$

Which can be discretized as (see Lecture 2)

$$\rho(t + \Delta t, x) = \rho(t, x) + \frac{\Delta t D}{\Delta x^2}(\rho(t, x-1) - 2\rho(t, x) + \rho(t, x+1))$$

to be compared with

$$p(t+1, x) = p(t, x-1)W_+ + p(t, x)W_0 + p(t, x+1)W_-$$

In order to have $p = \rho$, one need $W_+ = W_- = \Delta t D/(\Delta x)^2$ and $W_0 = 1 - 2\Delta t D/(\Delta x)^2 = 1 - W_+ - W_-$,and thus $\Delta t D/(\Delta x)^2 \leq 1/2$

Markov-Chain simulation of Diffusion

- Therefore a random walk is a way to sample a density ρ that obeys the diffusion equation.
    - With a random walk, it is easy to add obstacles, or aggregation processes, hard to include in the differential equation

More general case: Master equation

The probability to find the random exploration at location x at time $t+1$ is given by

$$
\begin{aligned}
p(t+1,x) &= \sum_{x'} p(t,x')W_{x'\to x} \\
&= \sum_{x'\neq x} p(t,x')W_{x'\to x} + p(t,x)W_{x\to x} \\
&= \sum_{x'\neq x} p(t,x')W_{x'\to x} + p(t,x)(1-\sum_{x'\neq x} W_{x\to x'}) \\
&= p(t,x) + \sum_{x'\neq x} [p(t,x')W_{x'\to x} - p(t,x)W_{x\to x'}]
\end{aligned}
$$

Detailed balance

In a steady state, the condition $p(x) = \rho(x)$ requires that

$$\sum_{x' \neq x} [\rho(x')W_{x' \to x} - \rho(x)W_{x \to x'}] = 0$$

We can choose $W_{x \to x'}$ according to the **detailed balance** condition

$$\rho(x')W_{x' \to x} - \rho(x)W_{x \to x'} = 0$$

Metropolis rule

Let us consider a physical system at equilibrium whose probability to be in state x is given by the Maxwell-Boltzmann distribution

$$\rho(x) = \Gamma \exp(-E(x)/kT)$$

We can sample this distribution with a stochastic process by choosing $W_{x\to x'}$ according to the **Metropolis rule**:

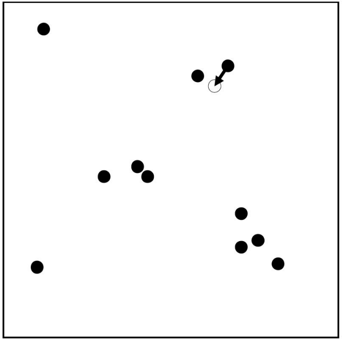
$$W_{x\to x'} = \begin{cases} 1 & \text{si } E' < E \\ \exp[-(E' - E)/kT] & \text{si } E' > E \end{cases}$$

The Metropolis Rule in practice

- In a gas, selects one particle at random
- Moves it by an amount Δx
- Computes the energy E' of the gas with this new position
- Accepts this change if

$$\mathrm{rand}(0,1) < \min(1, \exp[-(E' - E)/kT])$$

- By sampling ρ with $W_{x \to x'}$ can compute average physical properties, such as for instance the pressure in the gas



The Metropolis obeys the detailed balance

Let us assume that $E' > E$. Detailed balance is obeyed because

$$\begin{aligned}
\rho(x) W_{x \to x'} &= \Gamma \exp(-E/kT) \exp[-(E' - E)/kT] \\
&= \Gamma \exp(-E'/kT) \\
&= \rho(x') \times 1 \\
&= \rho(x') W_{x' \to x}
\end{aligned}$$

And similarly if $E' \leq E$

7. Monte-Carlo methods III

Kinetic/Dynamic Monte-Carlo

Let us consider the chemical equations

$$A \xrightarrow{k_1} B \qquad B \xrightarrow{k_2} A$$

They can be written as an ordinary equation

$$\frac{d}{dt}\begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} -k_1 & k_2 \\ k_1 & -k_2 \end{pmatrix}\begin{pmatrix} A \\ B \end{pmatrix}$$

Analytical solution

$$A(t) = \frac{k_2}{k_1 + k_2}(A_0 + B_0) + \frac{A_0 k_1 - B_0 k_2}{k_1 + k_2} e^{-(k_1+k_2)t}$$

$$B(t) = \frac{k_1}{k_1 + k_2}(A_0 + B_0) - \frac{A_0 k_1 - B_0 k_2}{k_1 + k_2} e^{-(k_1+k_2)t}$$

where A_0 and B_0 are the initial concentration of A and B. When $t \to \infty$

$$A \to A_\infty = \frac{k_2}{k_1 + k_2}(A_0 + B_0) \qquad B \to B_\infty = \frac{k_1}{k_1 + k_2}(A_0 + B_0)$$

Monte-Carlo Simulation

1. Defines a time step Δt, small enough so that $k_1\Delta t$ and $k_2\Delta t$ are smaller than 1. They are the **probabilities** that, during Δt, one A particle get transformed into one B particle, or conversely.

2. Chooses randomly a particle among the N = A(t) + B(t) = const of them. (In practice chooses a A particle rand(0,1) < A / (A+B), and a B particle otherwise.

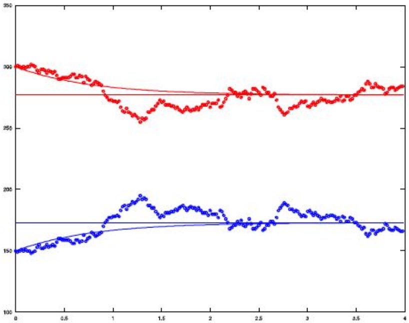
3a. If a A particle was chosen, it is transformed into a B particle provided rand(0,1) < $k_1\Delta t$. Then A = A - 1, B = B + 1

3b. If a B particle was chosen, it is transformed into a A particle provided rand(0,1) < $k_2\Delta t$. Then A = A + 1, B = B + 1

4. (2) and (3) are repeated N times and the physical time t is incremented by Δt: $t = t + \Delta t$

5. Repeats (2) - (4) until $t = t_{max}$

Results



Δt = 0.02 and k1 = 05.5, k2 = 0.8

The Monte-Carlo simulation fluctuate around the analytical solution. We should average over several runs.

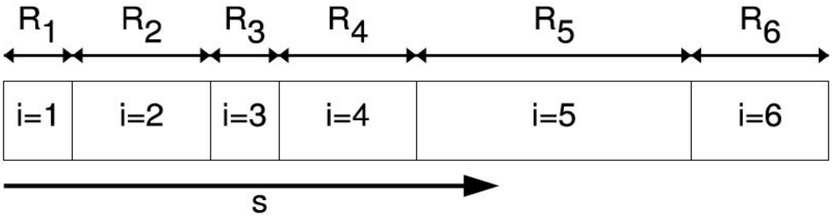
Gillespie's algorithm

1. Let r_i be the rate at which the possible events occur in the system. $i = 1,...n$.

2. For instance, $r_i = kAB$ for a reaction A + B → C

3. Let $R_i = \sum_{j=1}^{i} r_j$ be a cumulative rates.

4. Choose one event k at random by picking a random number s = rand(0,1). Chooses the k which verifies $r_{k-1} < sR_n < R_{k+1}$ (probability proportional rate).



Gillespie's algorithm

5. Execute the selected event (for instance a molecule A and molecule B disappear to produce a new molecule C).
6. Advance time as $\Delta t = \ln(1/s')R_n^{-1}$, with s' = rand(0,1).
7. Note that here Δt is calculated according to a decreasing exponential distribution. It gives the average time of occurrence of the next event.
8. Only one event takes place during the time interval Δt.